

Presentation - Matgeo

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Problem Statement

Find the area of the triangle whose vertices are $(-1, 1)$, $(0, 5)$, and $(3, 2)$

Solution

The area of the triangle formed by three point vectors **A**, **B** and **C** is

$$area(ABC) = \frac{\|(\mathbf{B} - \mathbf{A}) \times (\mathbf{C} - \mathbf{A})\|}{2} \quad (0.1)$$

Here,

$$\mathbf{A} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \quad (0.2)$$

$$(\mathbf{B} - \mathbf{A}) = \begin{pmatrix} 1 \\ 4 \end{pmatrix}, (\mathbf{C} - \mathbf{A}) = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \quad (0.3)$$

$$\|(\mathbf{B} - \mathbf{A}) \times (\mathbf{C} - \mathbf{A})\| = 17 \quad (0.4)$$

Therefore,

$$area(ABC) = \frac{17}{2} \quad (0.5)$$

Plot

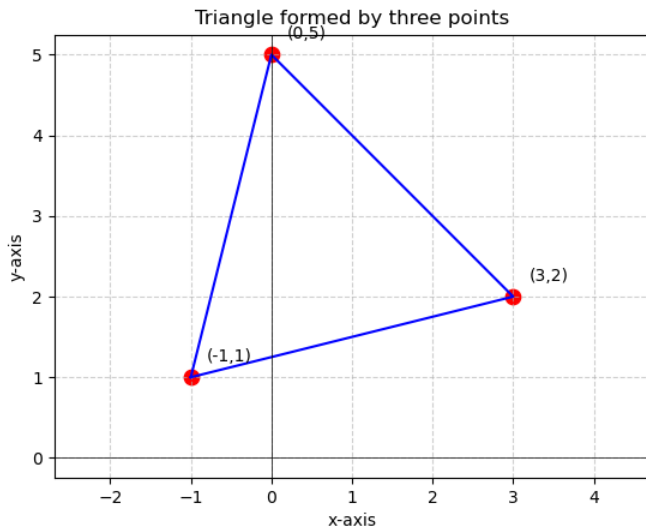


Figure: The Plot